

MOCOF AI Quotation Converter

Project Requirements, Implementation Status, Operations Guide and Development Handover

Prepared for MOCOF SDN BHD | 24 August 2026

Purpose

This handover records the intended product, the implemented codebase, known limitations, deployment configuration, and the next work required. It is written so the project can be continued without relying on the previous account or chat history.

Repository: github.com/Chen-maker-pen/mocof-ai-quotation-converter

1. Product goal

MOCOF needs a conversion system that takes a Chinese supplier renovation quotation and produces a customer-ready, editable English/MYR quotation. The customer version must follow the MOCOF Google-Sheets-style format, preserve supplier evidence and product photos, support manager edits and AI commands, then export to Excel or PDF.

Required workflow

- Upload a raw Chinese supplier XLSX or PDF quotation.
- Identify the quotation Area from the number of real customer rooms only. Services such as Extra m2, Curve, Wall Panel, Aluminium Frame and other add-ons are not rooms.
- Select the matching Area 1-10 prompt recipe and apply its required changes in sequence to the worksheet cells.
- Show one complete editable customer workbook: header, Whole House Total, Supplementary, and all room/detail tables required by that Area.
- Keep original data/photos for audit; translate customer-facing descriptions to English or bilingual text where required.
- Allow direct cell editing and manager prompt commands with an audit trail.
- Review totals, discounts, formulas and customer details before XLSX/PDF export.

Non-negotiable business rules

The supplied MOCOF Prompt Documentation is the source of truth. Fang and Joyce quotations are examples for validation only; their prices and layouts must not become hard-coded rules for future quotations.

2. Current implementation

Component	Current state
React + Vite frontend	Implemented. Upload page, workbook grid, direct cell editing, prompt drawer/trace, save draft controls and export controls are present.
XLSX supplier parser	Implemented with ExcelJS. Reads sheets, rooms, line items and embedded image references where available. Uses supplier summary totals as a safe baseline.
PDF supplier parser	Implemented through Gemini extraction. Requires GEMINI_API_KEY and is more sensitive to API/runtime time limits.
Area detection	Implemented. Counts real rooms and excludes standard service/add-on rows.
Area prompt storage	Implemented. All 415 prompt entries across Areas 1-10 are preserved in server/documentedPrompts.ts and shown in the prompt trace.
Spreadsheet grid	Implemented as A:J addressable grid with fixed H/I/J pricing columns so cell references can be reviewed.
Manager prompt transactions	Implemented. Sends boss-entered instructions to Gemini and applies returned cell transactions with audit status. Requires Gemini API key.
XLSX / PDF export	Implemented using ExcelJS and jsPDF. Export needs project-level testing with real quotations before production approval.
Vercel deployment	Configured via api/index.ts and vercel.json. Upload conversion has previously encountered function timeouts.

Important accuracy statement

The repository stores, traces and can send the full Area prompt sets, but it does not yet contain a complete deterministic line-by-line executor for every documented cell instruction across all Areas. The generic grid and selected AI operations are not sufficient to guarantee a customer quotation matches the boss's manual Google Sheets result. This is the main remaining engineering task.

3. Architecture and files

Location	Responsibility
src/	React UI: upload flow, editable workbook grid, prompt UI and quotation review.
server.ts	Express server and API routes for upload, convert, prompt application, save, approval and export.
api/index.ts	Vercel serverless entry point and route rewrite handler.
server/xlsxParser.ts	Supplier XLSX parsing, area detection and source-shape preservation.
server/documentedPrompts.ts	Auto-generated copy of the full MOCOF Prompt Documentation: exact Area 1-10 prompts. Do not hand-edit.
server/customerWorkbookGrid.ts	Builds the current editable A:J customer workbook grid.
server/calcEngine.ts	Uses integer cents/sen and preserves supplier room summary totals unless an explicit rule/command changes them.
server/geminiService.ts	Gemini extraction, translation and workbook prompt transactions.
server/exporter.ts	Customer XLSX/PDF export.
vercel.json	Vercel routes and 60-second API function duration.

Conversion data flow

Supplier XLSX/PDF -> parser -> Area detection -> documented recipe selection -> customer workbook grid -> manager review / prompt transactions -> save version -> XLSX or PDF export.

4. Prompt-document inventory

The complete prompt document was transferred into server/documentedPrompts.ts. The following table is an inventory; the actual prompt wording and cell coordinates remain in that source file for developers to execute in order.

Area	Documented recipe label	Prompt entries
1	Area 1 - 1 real room - 43 exact documented prompts	43
2	Area 2 - 2 real rooms - 42 exact documented prompts	42
3	Area 3 - 3 real rooms - 39 exact documented prompts	39
4	Area 4 - 4 real rooms - 41 exact documented prompts	41
5	Area 5 - 5 real rooms - 42 exact documented prompts	42
6	Area 6 - 6 real rooms - 42 exact documented prompts	42
7	Area 7 - 7 real rooms - 42 exact documented prompts	42
8	Area 8 - 8 real rooms - 40 exact documented prompts	40
9	Area 9 - 9 real rooms - 42 exact documented prompts	42
10	Area 10 - 10 real rooms - 42 exact documented prompts	42

How the prompt recipes must be executed

- Choose only the detected Area recipe. Do not mix Area 3, Area 7 or any other recipe.
- Start from the supplier workbook or a faithfully reconstructed workbook that retains the documented A1 coordinates.
- Run each documented prompt in its listed order. Create formulas, merges, labels, service rows and table sections at the coordinates specified by that Area.
- Calculate deterministically in the server. Gemini may translate/extract text and propose manager edits, but must not invent prices, quantities, discounts or formulas.
- Produce a transaction log: prompt number, affected cells, before value, after value, formula, confidence and review state.
- If a source layout conflicts with a recipe coordinate, preserve source evidence and raise a review exception instead of guessing.

5. Pricing, formula and quality controls

Customer quotations must be reviewable and mathematically defensible. The current calculation engine uses integer cents/sen to avoid floating point errors and no longer applies an invented default discount. Supplier room summary values are preserved as reconciliation baselines because merged source rows can otherwise be double-counted.

Controls required before sending a quotation

- Verify detected Area and real room count.
- Confirm customer name, address, sqft, budget, currency/exchange rate and discount cells.
- Confirm Whole House Total includes real rooms followed by add-ons/services.
- Confirm Supplementary has all Area-required rows, sqft/per, quantity, package prices, before price, after price and total rows.
- Confirm every room has its required Cabinet, Accessories, Wall Panel, LF, Kitchen/Vanity or other detail tables.
- Confirm all pricing formulas, total rows and cheapest-price highlight against the documented recipe.
- Confirm product images are retained in the exported XLSX when they exist in source.
- Manager approves the final output before it is sent to the customer.

Known issues / risks

Risk	Required action
Full recipe execution is incomplete	Build an AreaRecipeExecutor with a formula/table template for each Area and automated tests against approved expected output workbooks.
Vercel conversion can time out	Keep AI calls short; parse XLSX locally/deterministically first; use async job/storage architecture for larger PDFs and workbooks.
Gemini translation disabled by default	MOCOF_ENABLE_GEMINI_TRANSLATION must be true only after API cost, timeout and accuracy tests are approved.
Source formatting may differ	Preserve the original XLSX worksheet/template instead of generating generic grid-only output when exact Google Sheets formatting is required.
PDF export failures	Add automated export tests and verify jsPDF output for long worksheets, Unicode/Chinese text and embedded images.
Videos of manual steps	The supplied recording could not be decoded by the local analysis runtime. Use a standard H.264/AAC MP4, audio transcript and selected screenshots for workflow extraction.

6. Local development and deployment

Local setup

```
git clone https://github.com/Chen-maker-pen/mocof-ai-quotation-converter.git
cd mocof-ai-quotation-converter
bun install
GEMINI_API_KEY="your_key" bun run dev
```

Open **http://localhost:3000**. The project uses Bun, React, TypeScript, Vite, Express, ExcelJS, jsPDF and the Gemini API.

Vercel setup

- Import the GitHub repository into Vercel.
- Set GEMINI_API_KEY in Project Settings -> Environment Variables for Production and Preview.
- Do not commit API keys or .env files to GitHub.
- Redeploy after changing environment variables.
- Vercel currently routes /api/* through api/index.ts and allows 60 seconds per configured API function.
- For larger PDF/XLSX files or long AI extraction calls, move conversion to an asynchronous worker/storage system rather than relying on a single serverless request.

Recommended acceptance test matrix

Test case	Expected result
Area 3 XLSX	Three actual room rows; Whole House Total, Supplementary and Master/Living/Vanity detail tables appear in correct sequence.
Area 7 XLSX	Seven actual room rows; Area 7 prompt recipe used with no hard-coded Fang/Joyce values.
Supplementary	Prompt-defined rows/formulas and total are displayed and match approved Google Sheets calculations.
Manual prompt	Adding, editing or deleting a boss prompt creates a visible, auditable cell transaction and recalculates affected totals.
Export	XLSX opens in Excel/Google Sheets; PDF has no clipped tables, missing Unicode text or wrong totals.

7. Recent change log

Commit	Summary
6f05cc5	Preserve supplier summary totals during conversion to avoid double-counting merged line items.
fadc643	Preserve source prices until a specific recipe or manager action explicitly changes them.
239bda1	Remove sample-specific Fang/Joyce conversion recipes to avoid hard-coding examples as production rules.
29daa6b	Add early Area 3 / Area 7 conversion recipe work; superseded as a universal price/pattern approach.
82e930c	Replace generic README with MOCOF product overview.

Release / handover checklist

- Keep the GitHub repository private until customer data, API keys and commercial rules have been reviewed.
- Share this PDF with the next owner/developer together with the original MOCOF Prompt Documentation.docx and approved source-output quotation pairs.
- Give the next owner access to the GitHub repository, Vercel project and Gemini API billing project separately. Do not transfer passwords in this document.
- Before going live, complete deterministic AreaRecipeExecutor implementation and acceptance tests for Areas 1-10.
- Do not treat any generated quotation as customer-ready until a MOCOF authorised manager approves it.

Source records

Primary requirements source: MOCOF Prompt Documentation.docx (local source supplied to the project).
Generated in-code recipe source: server/documentedPrompts.ts. Example quotations/videos are validation materials only and are not pricing templates.